a ACPU-1 in the system

Host CPU
PCIe 6.0 root complex

▼

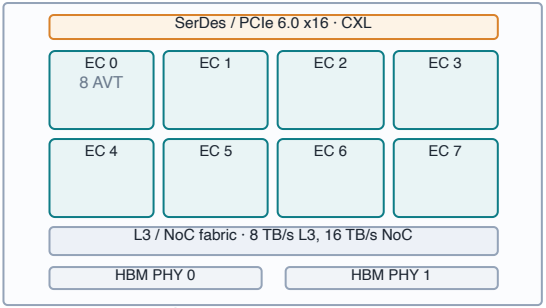
ACPU-1 card · 65 W TDP · dual slot
compute die 287.6 mm² · 2 × 64 MB stacked L3 · 2 × 24 GB HBM3E

▼

system DRAM · NVMe · NIC

x16 link: 121 GB/s · the 25.8 GB model state is loaded once per forecast, then streamed on-card.

b Compute die — logical floorplan



18.2 × 15.8 mm · 28.4 G transistors · 8 ensemble clusters

c Memory path of one time step

level	size	bandwidth	pJ / byte
HBM3E	48 GB	2.4 TB/s	16.0
L3 (stacked)	128 MB	8.0 TB/s	0.40
L2 per cluster	3 MB	1.2 TB/s	0.138
AVT SRAM	256 KB	550 GB/s	0.075
register file	32 KB	—	–

Each cell's 24 variables are read from HBM once and written once per step; 51.5 GB per step at 2.4 TB/s = 21.5 ms, overlapped with 29.1 ms of compute.

d Where a time step goes

time — 29.1 ms total



compute 21.6 ms — memory 21.5 ms (overlapped)

energy — 1.26 J total



compute 26% · HBM 56% · static 15%

Arithmetic is 26% of the energy. The design is paid for in bytes, which is why the fabric and the fused pass matter more than peak FLOPs.